

Per-Capability Quantization Degradation on Blackwell

A Throughput–Retention Study

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github.com/vshortt73/blackwell-quant-tradeoff

Status: first complete three-arm sweep. The serving and perplexity results are final. The per-capability (APEX) result is **inconclusive and reported as such**; Section 4.3 states why, and Section 6 states what would change it.

Abstract

As quantization drops BF16 → FP8-native → AWQ-4bit on an RTX 5090 (sm_120), throughput rises — but not monotonically, and not in the same order at every concurrency. Measured under locked clocks with all other server flags held identical, AWQ-4bit is **2.2× BF16 at concurrency 1** yet **FP8-native overtakes it at concurrency 64** (3,176 vs 2,639 output tok/s), because the regime shifts from weight-bound to compute-bound and native FP8 tensor cores beat dequantize-then-compute. On quality, paired per-passage perplexity over 82,282 held-out tokens finds FP8 **indistinguishable from BF16** (ratio 1.0011, 95% CI [0.9997, 1.0024]) while AWQ-4bit costs **6.0%** (ratio 1.0601, CI [1.0548, 1.0657]). The decision-relevant statement for a serving team is therefore: *FP8-native buys 44% more throughput at high concurrency for no measurable quality cost*. The intended headline — whether the three APEX capability dimensions degrade non-uniformly — could not be answered on this model: only one of three dimensions exhibits a positional curve at BF16, and bootstrapped curve-strength intervals overlap across all three schemes.

1. Motivation

A serving team choosing a quantization scheme needs more than “quality dropped X%”. It needs to know *which* capability drops and at *what* precision, because an agent workload that depends on instruction-following has a different tolerance than a RAG workload that depends on recall.

Published vLLM benchmarks report throughput, and sometimes a single aggregate quality number. None report a per-capability degradation profile. This study set out to produce one, using APEX’s three independent dimensions — factual recall, instruction following, salience — as the quality instrument, anchored by task accuracy and held-out perplexity.

It delivers the throughput–quality tradeoff with tight intervals. It does not deliver the per-capability profile, for reasons that are themselves a result (Section 4.3).

2. Background

Blackwell sm_120 serving. The RTX 5090 is consumer Blackwell. As of torch 2.13.0+cu130 and vLLM 0.27.1, a stock `pip install vllm` produces a working sm_120 stack with no special index or source build — the early-window claim that vanilla installs reject sm_120 does not reproduce. Native FP8 is available on Linux via `CutlassFp8BlockScaledMMKernel`; the emulated fallback is a WSL2/dxgkrnl artifact and does not apply. Both were verified rather than assumed, and both are recorded in `LANDMINES.md` alongside two failures that do still bite (Section 3).

APEX measures position-influence curves: a probe is embedded at a controlled position within a filled context, and a downstream query measures whether the probe influenced the answer. It reports per-position scores; it does not report whether a curve *exists*. That verdict is derived here (Section 3).

Positional salience. The U-shaped “lost in the middle” effect (Liu et al., 2024) is the canonical positional phenomenon this instrument is built to detect, and it is what the one measurable dimension in this study exhibits.

3. Method

Full pre-registration in `METHODLOGY.md`; only the parts that materially shaped the result are restated here.

Experimental matrix

Factor	Specification
Independent variable	quantization scheme: BF16 / FP8-native / AWQ-4bit
Model	Qwen3-8B, three official checkpoints from the same base weights
Controlled	GPU, locked SM clock (set 2400 MHz, observed 2377–2392), power limit (500 W), gpu_memory_utilization 0.85, max_model_len 12288, attention backend, DeepGEMM off, ignore_eos, greedy decoding, prompt set, probe set, evaluator
Dependent	output-token and request throughput; TTFT/TPOT/E2E at p50/p90/p99; energy; task accuracy; held-out perplexity; per-dimension curve existence and strength

The full-precision arm is **bfloat16**, the checkpoint’s native dtype. Forcing float16 on a bf16-trained model would degrade the reference point every other arm is measured against.

Model size is hardware-bound. A three-arm sweep on one 32 GiB card caps the model at ~10B: the BF16 arm alone needs 15.3 GiB. This constraint is load-bearing for the negative result in Section 4.3.

Two configuration errors would have silently corrupted the result, and are recorded in `LANDMINES.md` because neither announces itself:

1. *DeepGEMM on sm_120.* vLLM defaults `VLLM_USE_DEEP_GEMM=True`, and its scale-factor layout transform aborts loading block-scaled FP8 weights (“Unknown SF transformation”). `VLLM_USE_DEEP_GEMM=0` is required, and is set for **every** arm so the environment stays identical.
2. *Qwen3 reasoning tokens.* Qwen3 emits `<think>` blocks by default and APEX’s scorers strip only whitespace, so programmatic probes score the reasoning text rather than the answer — measured at 0.332 against 0.983 for exact-match. Both `--reasoning-parser qwen3` and `no_think: true` are required; neither suffices alone. Dangerously, a quantized model may emit a *different amount* of reasoning, so the artefact would read as damage to instruction-following.

Curve existence is derived, not reported by APEX

A dimension has an exploitable curve if position explains variance in score beyond chance, *after removing probe difficulty*:

- `curve_strength` = within-probe eta-squared (each probe’s mean subtracted, then `SS_between_positions / SS_total` on the residuals)
- `curve_exists` = seeded permutation test on that statistic, $p < 0.05$

Eta-squared rather than a correlation because positional effects are frequently **non-monotonic** — a U-shape scores ~0 on Pearson. Permutation rather than ANOVA’s F because scores are bounded in [0,1] and far from normal. Within-probe rather than pooled because, measured across 11,189 rows of prior APEX history, probe identity explains **0.630** of score variance against position’s **0.031** — pooling drowns the signal in difficulty differences. Re-analysed within-probe, significant cells in that history rise from 1/35 to 7/34.

Repetition explains **0.000** of variance, confirming greedy decoding is exactly deterministic. Repeats therefore carry no information about uncertainty for quality metrics; intervals come from bootstrapping over passages (perplexity) or probes (curve strength).

Probe-set calibration. Probe difficulty is relative to the model under test. `harness/check_probes.py` measured all 60 seed probes against BF16 and found only **28 usable**: 14 factual probes at ceiling, 14 salience probes at floor. The sweep runs the usable subset; the excluded 32 cannot register a result and would roughly double runtime.

Evaluator. Rubric-scored probes (all salience, 15/20 application) are graded by Qwen2.5-14B-Instruct-Q4_K_M on a separate host — never the model under test, which would make the judge degrade in lockstep with its subject. It was fingerprinted before and after the sweep; both reads were 98f010b1ff545412, so the judge did not move.

4. Results

Environment: RTX 5090, driver 580.159.03, CUDA 13.0, torch 2.13.0+cu130, vLLM 0.27.1, flashinfer 0.6.16.post3, sm_120, clocks locked at 2400 MHz (observed 2377–2392) / 500 W. All results stamped with this fingerprint and a clean git commit.

4.1 Throughput and latency

Output-token throughput (tok/s), 200 requests per level, 512-in/256-out, ignore_eos forcing exactly 256 decoded tokens per request in every arm:

Scheme	c = 1	c = 4	c = 16	c = 64
BF16	90.8	323.1	1032.6	2199.9
FP8-native	125.9	474.0	1497.8	3175.5
AWQ-4bit	202.1	709.2	1821.6	2638.8

The ordering inverts. AWQ leads by 2.2× at concurrency 1 and still leads at 16, but FP8 overtakes it at 64 by 20.3%. At low concurrency the workload is weight-bound and 4-bit weights win on memory traffic; at high concurrency it turns compute-bound, and native FP8 tensor cores beat dequantize-then-compute.

Latency and energy at the extremes:

Scheme	TTFT p50 (c=1)	TTFT p99 (c=64)	TPOT p50 (c=64)	tok/J (c=64)
BF16	50.5 ms	2420 ms	23.76 ms	5.52
FP8-native	34.2 ms	1501 ms	16.85 ms	8.16
AWQ-4bit	49.7 ms	2490 ms	21.00 ms	7.12

FP8 takes best time-to-first-token at every level, best tail latency under load by a wide margin (1.5 s vs ~2.4–2.5 s at p99), and best energy efficiency.

4.2 External quality anchors

Held-out perplexity over 198 private, unpublished passages (82,282 scored tokens), compared **paired per passage** — every arm scores the identical token sequence, so passage difficulty cancels:

Comparison	Perplexity	Ratio	95% CI	Verdict
FP8 vs BF16	36.451 → 36.490	1.0011	[0.9997, 1.0024]	not distinguishable from zero
AWQ vs BF16	36.451 → 38.642	1.0601	[1.0548, 1.0657]	significant

Pairing is what makes this resolvable. On the same data the unpaired interval is [−0.0667, +0.1833] nats/token — it spans zero and would report *no detectable difference* for a real 6% degradation.

The corpus scores 38.6 perplexity against 11.5 for formulaic text, which is direct evidence it is genuinely unseen rather than recalled.

Task accuracy could not resolve either comparison. BF16 and FP8 answered all 50 items identically (0 discordant pairs); AWQ produced 1 regression ($p = 1$, McNemar exact). At $n=50$ with a 0.92 baseline only ~3 items sit near the decision boundary, so flips are vanishingly rare. This was predicted by the task validator before the run, not discovered after it.

4.3 APEX per-dimension retention

This is the section the study was built for, and it does not deliver a finding.

Curve verdicts, within-probe permutation test:

Dimension	BF16	FP8-native	AWQ-4bit
factual_recall	0.112 (p=0.130)	0.067 (p=0.595)	0.141 (p=0.034)
instruction_following	0.085 (p=0.0002)	0.063 (p=0.010)	0.069 (p=0.004)
salience	0.097 (p=0.231)	0.048 (p=0.834)	0.061 (p=0.682)

Only instruction_following has a curve at BF16. Per the pre-registered outcome, a dimension with no curve at full precision is a property of the model and probe set, not of quantization, and is excluded from the degradation claim rather than reported as a collapse. That removes factual_recall and salience.

The curve that does exist is a clean **lost-in-the-middle U**, reproduced in all three arms — mean score by position, instruction_following:

Position	0.02	0.20	0.50	0.80	0.98
BF16	0.860	0.725	0.710	0.796	0.815
FP8	0.807	0.783	0.774	0.792	0.813
AWQ	0.807	0.697	0.727	0.732	0.779

Curve strength appears to fall from 0.085 to 0.063 under FP8. **It is not distinguishable from noise.** Bootstrapping over probes — the independent unit — gives heavily overlapping intervals:

Dimension	BF16	FP8	AWQ
factual_recall	[0.080, 0.396]	[0.054, 0.302]	[0.080, 0.445]
instruction_following	[0.048, 0.195]	[0.041, 0.152]	[0.045, 0.171]
salience	[0.080, 0.308]	[0.035, 0.295]	[0.061, 0.301]

The factual_recall cell that clears threshold under AWQ (p=0.034) should not be read as a finding: nine tests were run without correction, ~0.45 false positives are expected at $\alpha=0.05$, and it does not survive Bonferroni. Reading it as “quantization created a curve” would be exactly the kind of noise-mining the pre-registration exists to prevent.

With one measurable dimension and intervals this wide, H1 cannot be tested. Not rejected — untested. The instrument works; the measurement is underpowered.

5. Discussion

For a serving team, the recommendation is unambiguous. FP8-native is the default choice on Blackwell: 44% more throughput than BF16 at concurrency 64, best TTFT and tail latency at every level, best energy efficiency, and no measurable quality cost against an 82k-token held-out corpus. AWQ-4bit is the right choice only for low-concurrency, latency-insensitive, memory-constrained deployments where its 2.2× single-stream advantage matters more than 6% perplexity.

The regime inversion is the transferable methodological point. A benchmark publishing one throughput number would name AWQ or FP8 the winner purely according to which concurrency it happened to sample. Reporting single-stream and concurrent regimes separately is not fastidiousness; it is the difference between a correct and an incorrect recommendation.

Why the per-capability claim failed is itself informative. It was not a tooling failure — the detector found a textbook U as soon as it was given a dimension in range. It was a *calibration* failure, in two directions at once: factual probes are too easy for an 8B (14/20 at ceiling) and salience probes too subtle to move one (14/20 at floor). Probe sets have a difficulty, that difficulty is relative to the model, and a probe set validated on ~30B models does

not transfer down. The models that showed salience curves in prior APEX history were Qwen3-32B and qwen3-30b-a3b, both scoring mid-range where there is room to move.

That produces a hard constraint: curve detection wants mid-range baseline scores, which on this probe set means ~30B, while a three-arm precision sweep on one 32 GiB card caps the model at ~10B. **Those two requirements cannot both be satisfied on this hardware.** Resolving it requires either recalibrated probes for the 8B class, or a larger card.

6. Limitations

- **One model, one size, one family.** Qwen3-8B only. Cross-family generalization is not attempted.
- **The per-capability result is underpowered, not negative.** Six usable probes in two dimensions produce intervals too wide to compare across arms.
- **Two dimensions had no curve at baseline** and are excluded by pre-registration, leaving no basis to test non-uniformity.
- **Task accuracy is underpowered at n=50**, with 0–1 discordant pairs.
- **Single context length** (8,192) for the reported sweep. Prior history shows positional effects strengthening at 16k+, which was not sampled here.
- **The evaluator is an LLM** and enters the measurement chain. It was pinned, greedy, hosted separately, and fingerprinted identical before and after — a stable bias cancels in the deltas — but it is not a neutral instrument.
- **vLLM only**, one serving stack.
- **Publishing the task set eventually contaminates it**, as happened to GSM8K and MMLU. Reusers should re-author rather than inherit.

7. Future Work

In descending order of expected payoff:

3. **Author replacements for the 32 degenerate probes** — harder factual probes, stronger-signal salience probes, using the 28 survivors as templates. This directly narrows the intervals that made Section 4.3 inconclusive.
4. **Scale the task set to 300–1000 items**, so McNemar has discordant pairs to work with.
5. **Extend to 16k–32k context**, where prior history shows positional effects are strongest.
6. **Repeat on a ~30B model** where all three dimensions sit in range, accepting a two-arm (FP8 vs AWQ) comparison since a 30B BF16 arm does not fit 32 GiB.
7. Cross-family (Gemma, Llama); TensorRT-LLM cross-check; speculative decoding interaction.

Reproducibility

Everything needed to reproduce is committed: `env/requirements.lock` (203 pins from the validated stack), `env/Dockerfile`, `env/gpu_setup.sh` (clock lock), `configs/*.yaml` (controlled variables, one file per scheme), `env/run_sweep.sh` (the exact unattended sweep that produced these numbers), and `results/raw/*.json` (every result with a full environment fingerprint and the originating git commit). `data/prompts_512in.txt` and `data/tasks.jsonl` are committed; the held-out corpus is not, because publishing it would destroy the property that makes it a valid perplexity anchor.

Regenerate every figure and table from raw with:

```
python results/analysis.py
```

References

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